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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 12: EIA Processes - I

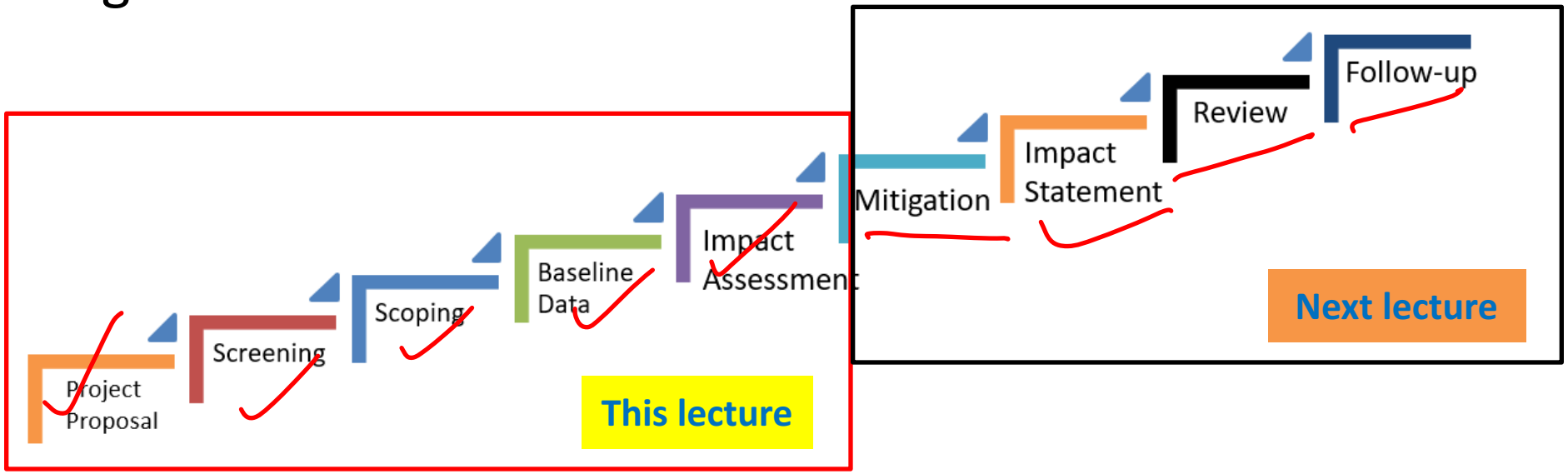
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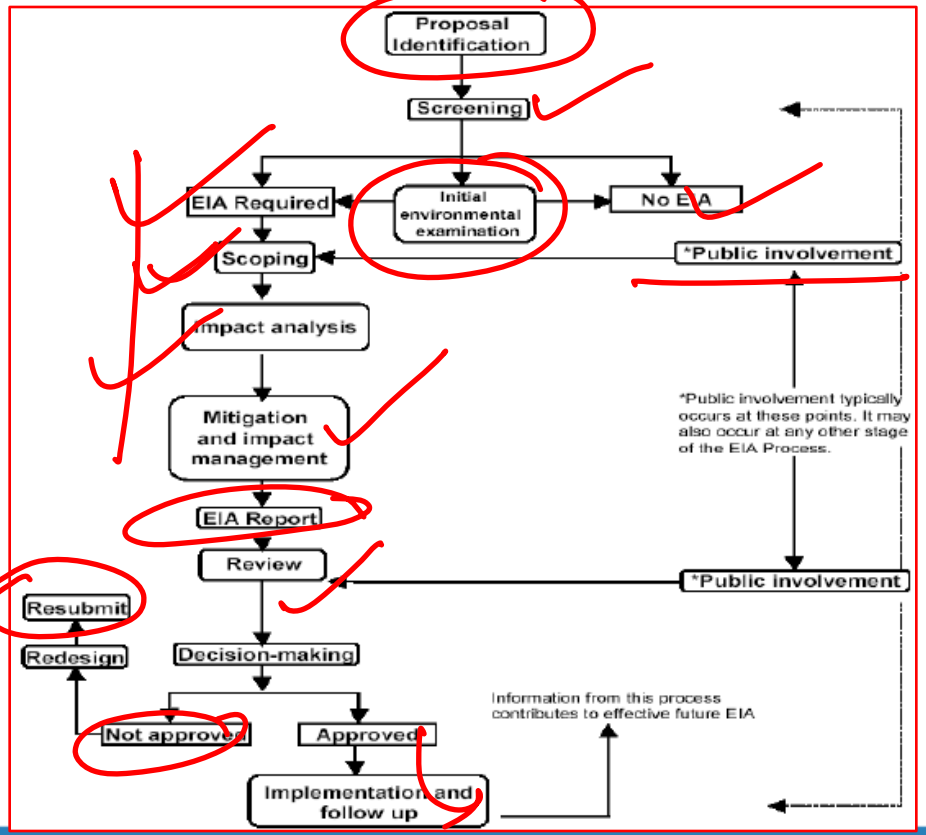
Contents

- Steps or stages of Environmental Impact Assessment process
- Project proposal in an EIA
- Screening process
- Scoping process
- Baseline study
- Impact assessment process

Stages of EIA

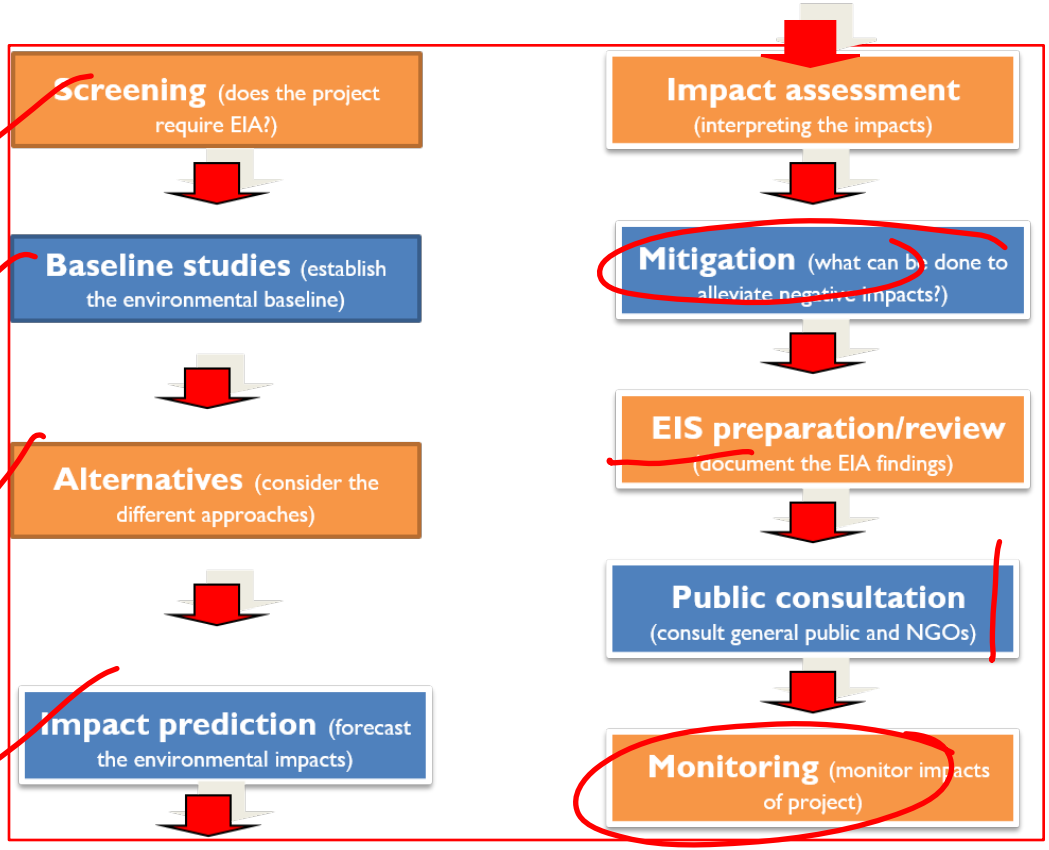


Flow Chart of Stages in EIA

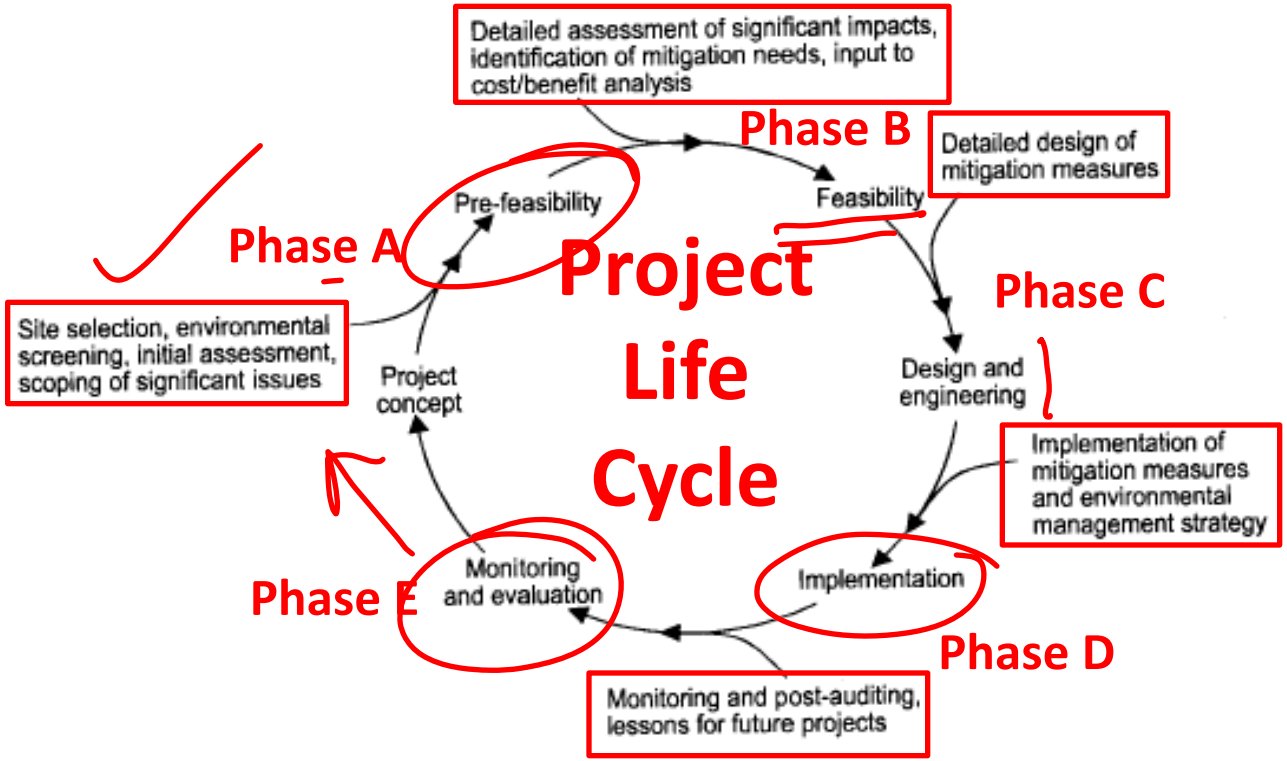


Source: (Dibya Jyoti Kalita, 2016)

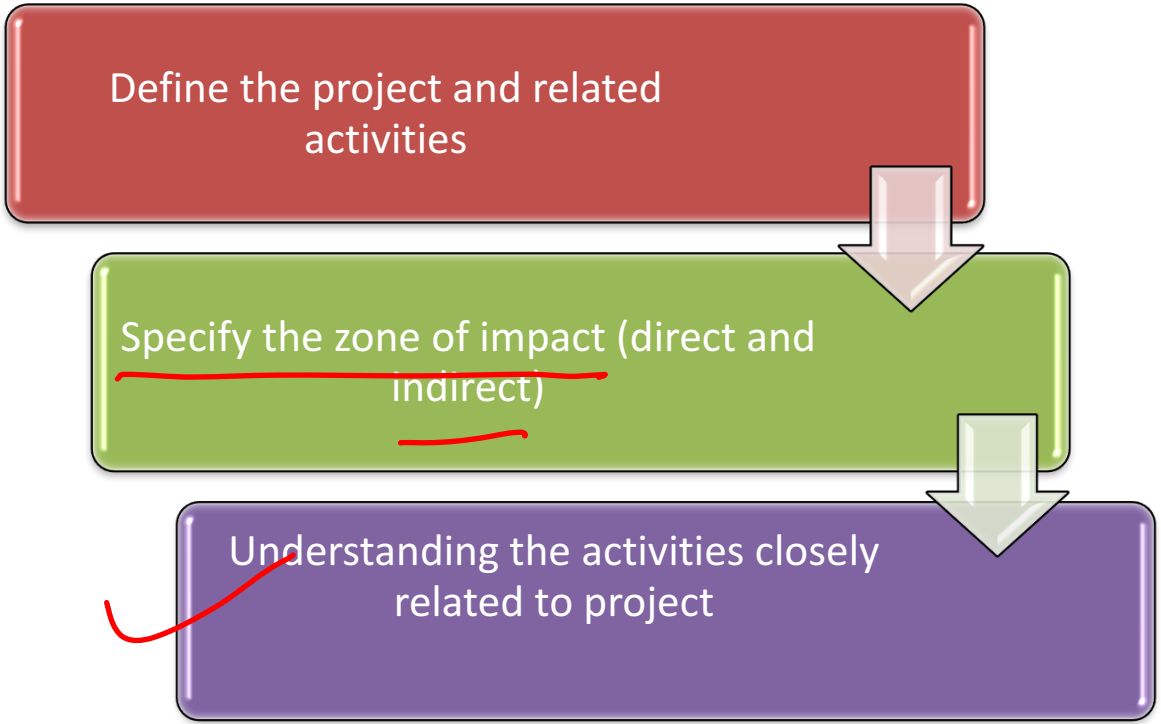
The EIA phases



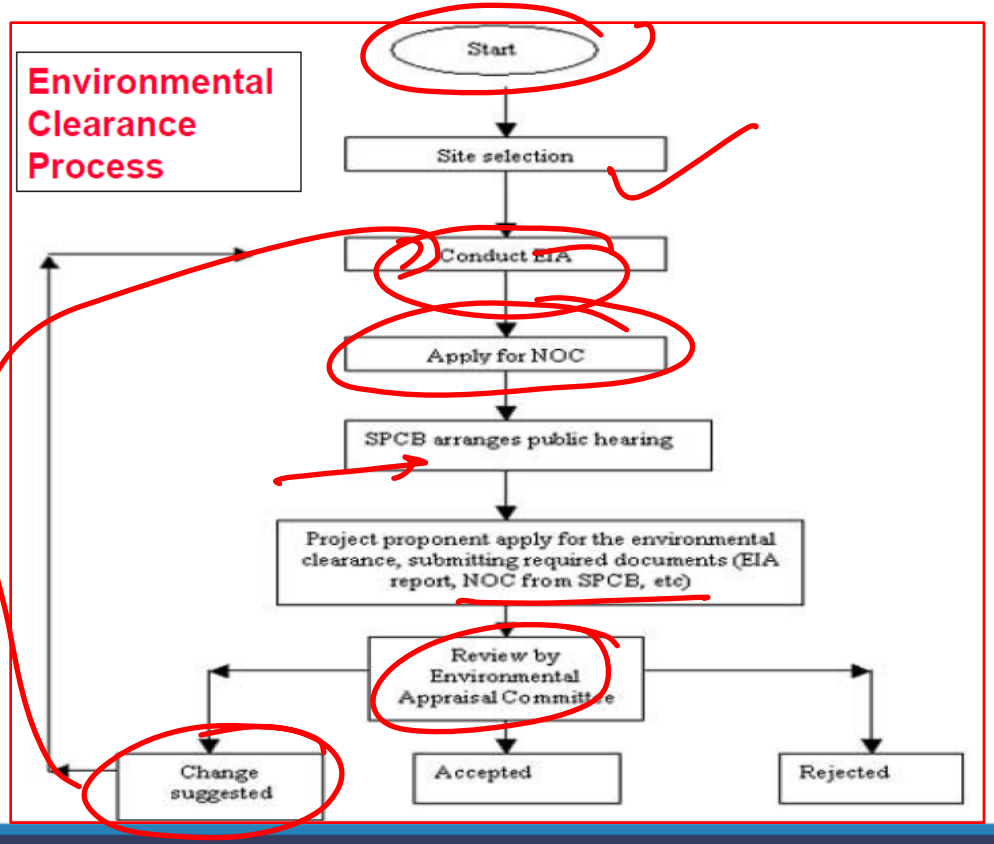
Example of EIA Phases explained through a Project Life Cycle



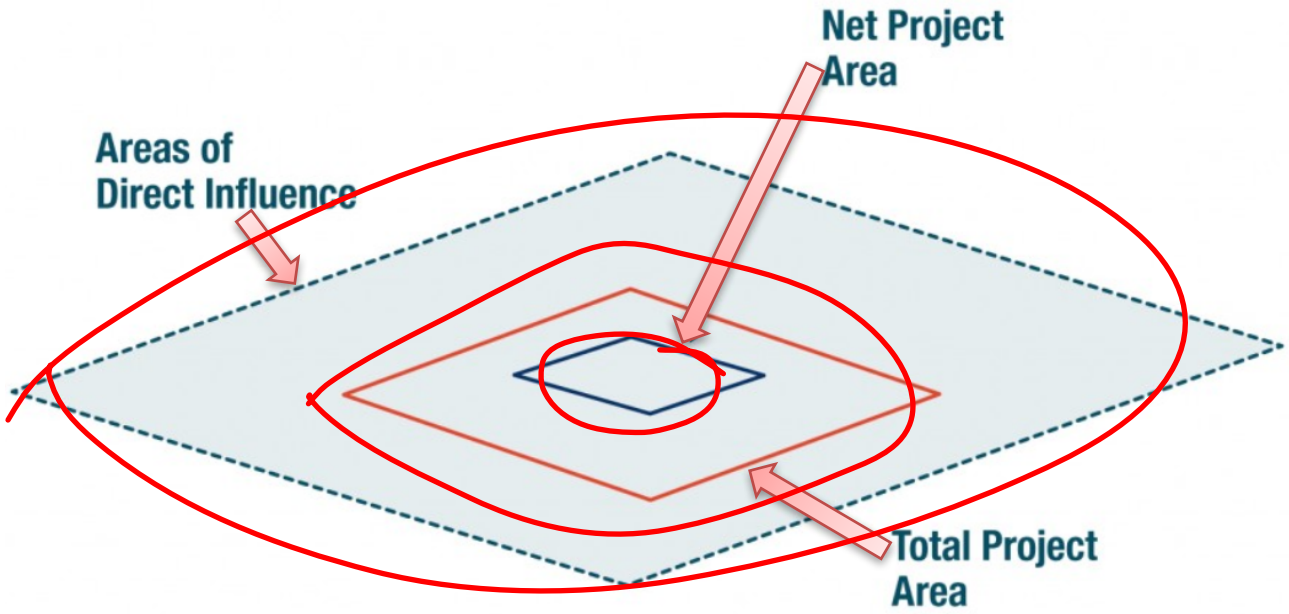
Project Proposal



Environmental Clearance Process

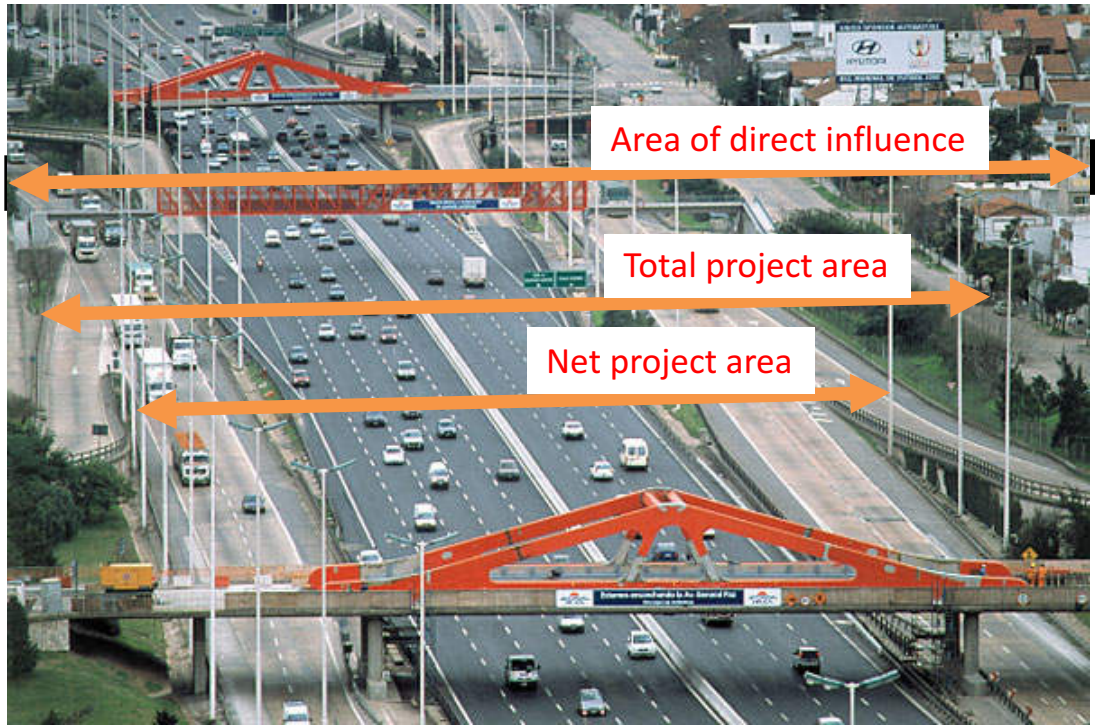


Zone of direct & Indirect Impact



Source: isd.org

Example of Zone of direct & Indirect Impacts



Example of a Highway

- Net project area: Right of way of highway.
- Total project area: Area up to buildings.
- Area of direct influence: Area affected by noise, air pollution or otherwise.

Screening process in EIA

To decide whether:

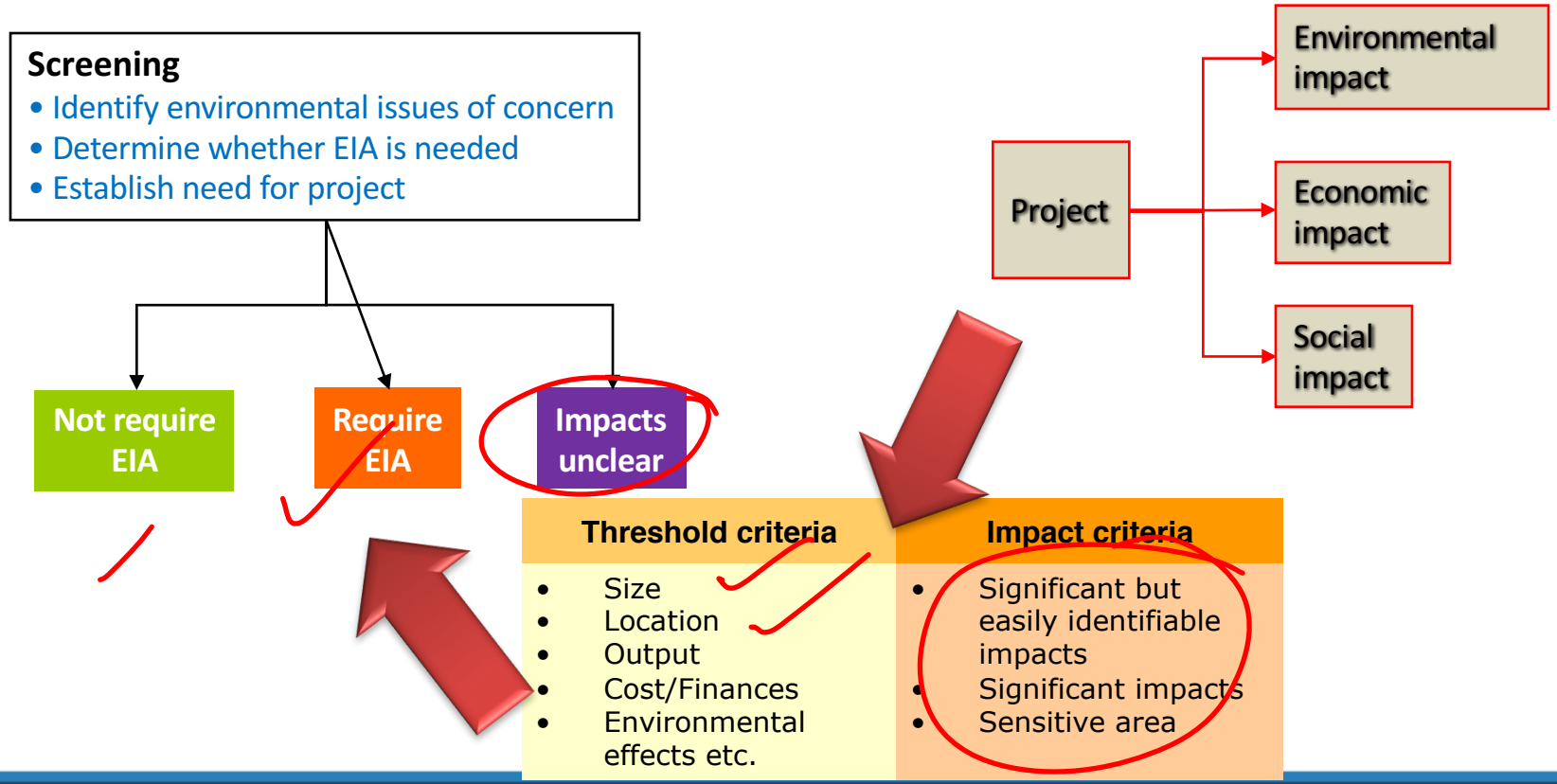
- To conduct the assessment (based on the likely significant impacts)

OR

- Not to conduct it (in the anticipated absence of such impacts).

Screening needs to follow specific procedures often described in the legislation (as per local laws).

Screening process flowchart



Why do we need screening?







An oil extraction or mining or a project of bullet train will have significant impacts

If screening was that simple, then why we needed lengthy processes such as EIA?

An Example: Why do we need screening?



Would you consider

- Increased vehicular traffic
- Increased air traffic at nearby airport
- More visitors and hospitality infra.
- Increased commercial traffic for supplies

To ease out the complex process, screening is done in a predetermined manner.

Let suppose a tourist project is proposed for Mussoorie, Uttarakhand. How will you do screening of this project?

Parameters in Screening Process

Type of Projects

Size of Project

Potential Impacts

Example.

- Type can be mining, river training, transport etc.
- Size of project can be classified as per cost, length of road or project area.
- Impacts can be long or short term, moderate or severe, reversible or irreversible etc.

Standardized Approach to Screening Process

1- Activities outlined in legislation and regulations.

2- Legal (or policy) definitions of proposals

3- Inclusion lists of projects (with or without thresholds)

4- Exclusion lists of activities that do not require an EIA because they are insignificant or are exempt by law (e.g., national security or emergency activities).

Example of Standardized Approach to Screening Process

1.1-All activities related to nuclear substances can be put in compulsory EIA category like Nuclear Power Plant

2-Any project which needs resettlement, e.g., a reservoir

1.3-Any four or more-lane highway project of more than 100 km length

1.4-Border infrastructure is exempted from EIA process in India

Customized Approach to Screening Process

- Individual or case-by-case base, using indicative guidance with categories.
- Different countries and international agencies combine these types of screening procedures.
- Most often, a simple categorization (such as A,B,C or 1,2,3) is used.
- These categories help proponents and licensing agencies decide when a full EIA is needed or not.

Customized Approach: For high-impact projects

- Projects likely to have significant serious adverse environmental impacts (result in irreversible damage, affect vulnerable ethnic minorities, involve involuntary resettlement, or affect cultural heritage sites).

Examples

- Industrial plants (large-scale).
- Reclamation, resettlement and new land development.
- Thermal and hydropower development;
- Project using pesticides and other hazardous and/or toxic materials.

Customized approach: For Medium-impact projects

- A limited EIA to identify suitable mitigation and management measures and incorporate them into the project.

Examples

- Electrical transmission-rural electrification
- Aquaculture, irrigation and drainage (small-scale)
- Renewable energy
- Tourism
- Rural water supply and sanitation

Customized approaches: For low-impact projects

- Project which are likely to have minimal or no adverse environmental impacts
- No EIA is required.

Examples

- Construction of small buildings
- Community garden development
- Development of wells in a community
- Outdoor recreation

Screening Approach: World Bank-A Category projects

Categories, depending on the type, location, sensitivity, and scale of the project and the nature and magnitude of its potential environmental impacts.

Screening Approach: World Bank-B Category projects

Potential adverse environmental impacts on human populations or environmentally important areas



Screening Approach: World Bank-C Category projects

A proposed project is likely to have minimal or no adverse environmental impacts.

Beyond screening, no further EA action is required.

Example

Small community project or streets or village roads under PMGSY (Pradhanmantri Gram Sadak Yojana)

Screening Approach: European Union (EU)

EIA depending on size, location, activities, and impacts.

- Power stations: capacity in MW
- Landfills: Total volume, volume/day, tones/day, total capacity in tones-
- Roads: length of road (in km)
- Pipelines for the transport of gas (length in km)







Image Source: bbc.com, thegurdian.com, krohne.com

EU Checklist for Screening: Yes/No or 1/2

1. Will there be a large change in environmental conditions?
2. Will new features be out-of-scale with the existing environment?
3. Will the effect be unusual in the area or particularly complex?
4. Will the effect extend over a large area?
5. Will there be any potential for transboundary impact?
6. Will many people be affected?
7. Will many receptors of other types (fauna and flora, businesses, facilities) be affected?
8. Will valuable or scarce features or resources be affected?
9. Is there a risk that environmental standards will be breached?

'Yes': EIA required whereas
'No': No EIA required.
In case of more 'Yes', EIA will be required.

EU Checklist for Screening: Yes/No or ½ (cont'd..)

11. Is there a risk that protected sites, areas, features will be affected?
12. Is there a high probability of the effect occurring?
13. Will the effect continue for a long time?
14. Will the effect be permanent rather than temporary?
15. Will the impact be continuous rather than intermittent?
16. If it is intermittent, will it be frequent rather than rare?
17. Will the impact be irreversible?
18. Will it be difficult to avoid, or reduce or repair or compensate for the effect?

Source: European Commission, 2001

Scoping process in EIA

- Scoping is a systematic exercise that establishes the boundaries of your EIA.
- It sets the basis of the analyses you will conduct at each stage.
- Setting the baseline and identifying alternatives.
- Reduces the risk of including inappropriate components or excluding components that should be addressed.

Area of study, factors for assessment for baseline studies are most important outcomes of scoping.

Example of Scoping process in EIA

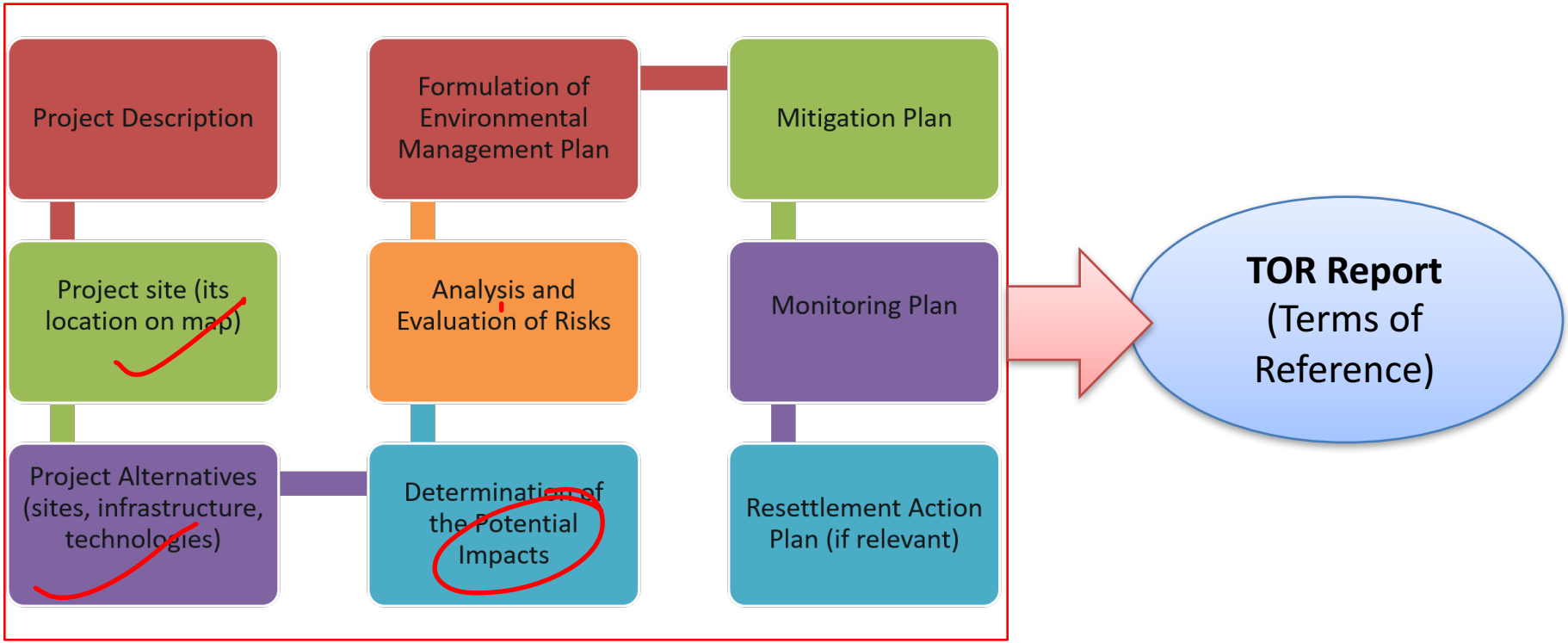


Image Source: citizenmatters.in

Char-Dham All Weather Road Project in state of Uttarakhand

- May have significant regional environmental, economic and social impacts which creates a very large scope boundary in case of EIA process.

Scoping Process Flowchart

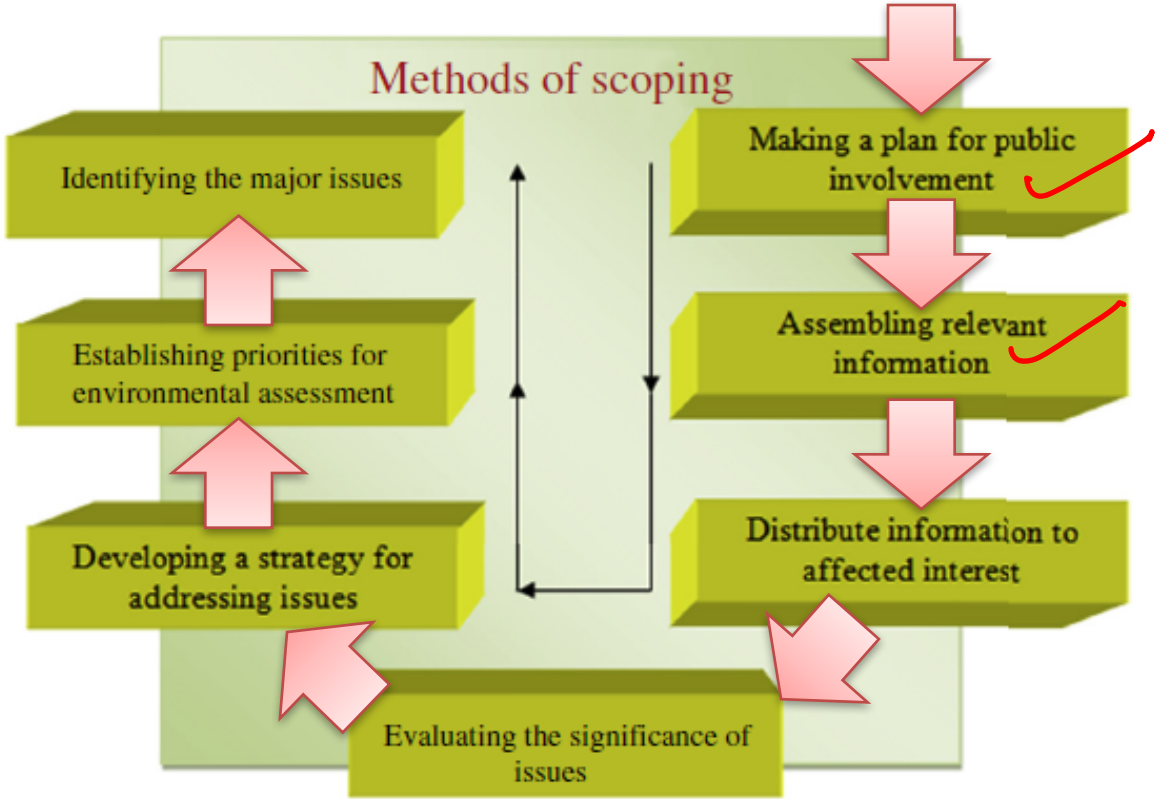


Terms of Reference (TOR)

- A description of the project
- A list of the agencies or ministries responsible for overseeing the EIA process and making decisions
- The geographic area to be studied (also called the 'impact zone')
- EIA requirements in applicable laws or regulations
- Impacts and issues to be studied
- Mitigation and/or monitoring systems to be designed
- Provisions for public involvement
- Key stakeholders
- Timeframe for completing the EIA process
- Expected work product and deliverables
- Budget for the EIA

The Terms of Reference (TOR) serve as a roadmap for EIA preparation and encompass the issues and impacts that have been identified during the scoping process.

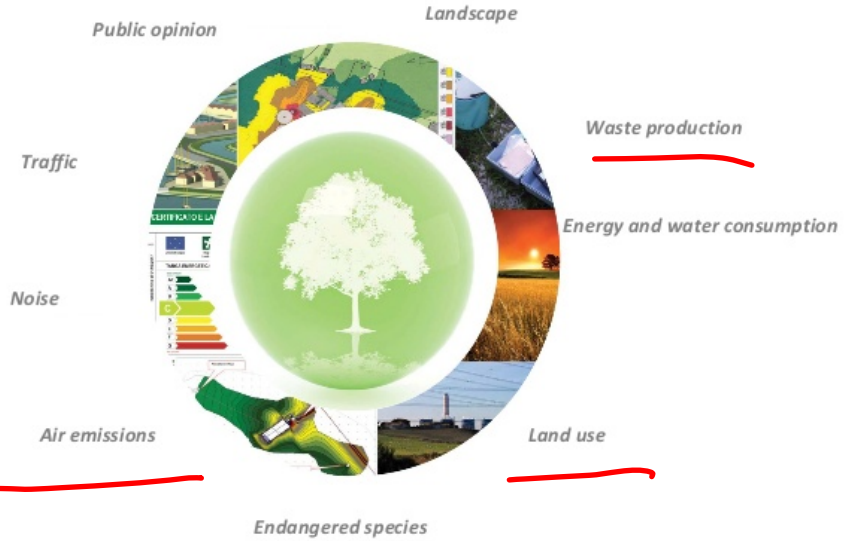
Methods of Scoping



Source: (Bai Yong and Bai Qiang, 2014)

Baseline Study in EIA

- Essential to collect all relevant information on the current status of the environment.
- Provide a sound basis for the design and evaluation of post-EIA studies
  - Ex. Early stage monitoring.
- Decide about a timeline that will allow for predictions of how the project may change the key environmental components
  - Ex. Number of years of project operations.



Impact Prediction

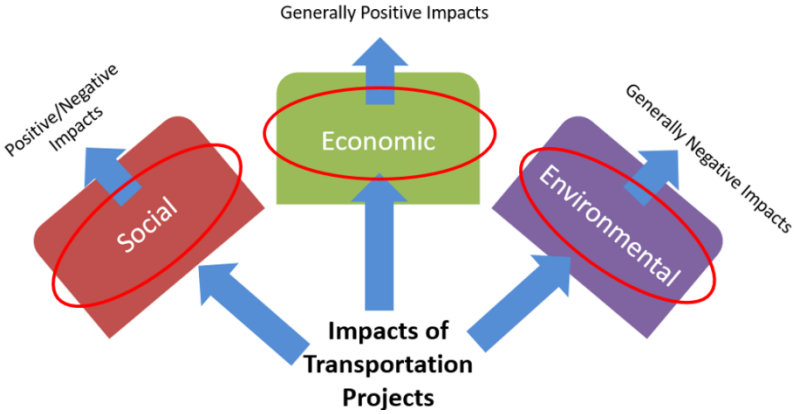
- Impact prediction involves forecasting the likely changes in the environment that will occur as a result of the development.

- Impacts of the project can change over time
  - Needs to be assessed and measured over the lifetime of the project: Construction phase to operations and even after closure.
    - Ex. Nuclear Power plants

Comparing the baseline data with various alternatives is done under impact prediction.

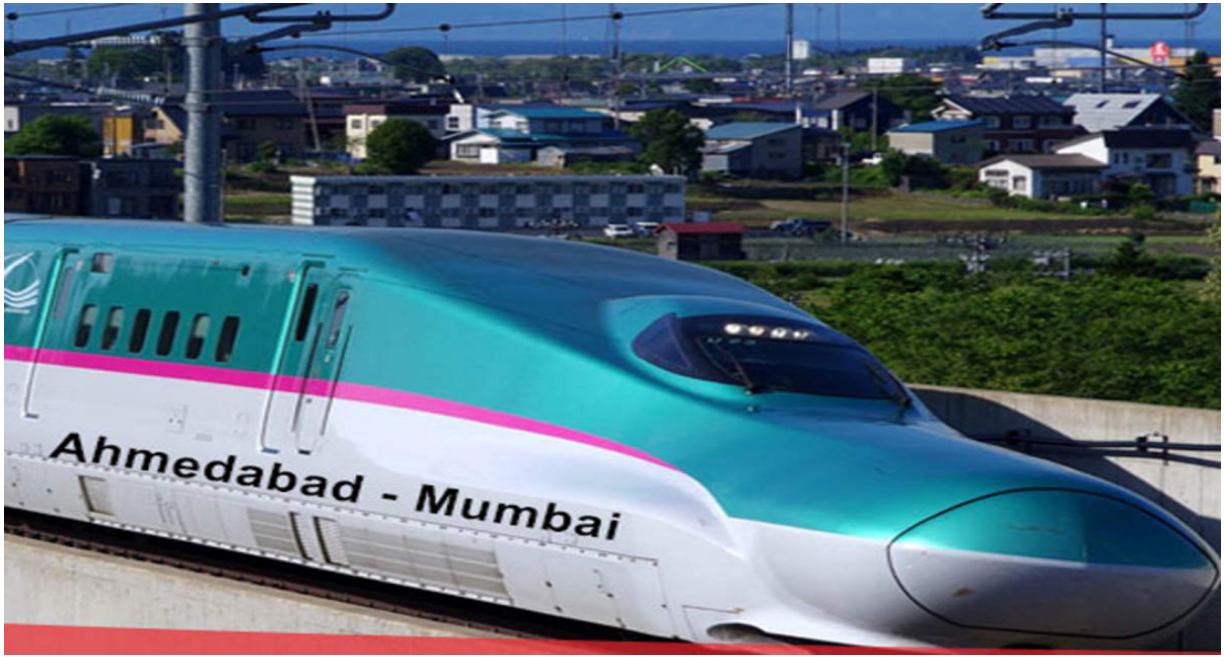
Impact assessment: Objectives and Approaches

- Based on the key environmental, social and cultural characteristics of the area.
- Provides a clear and itemized list of relevant impacts on the environment and people, including cumulative effects, social impacts, and health risks.
- Direct, indirect and cumulative impacts.
- Based on the results of the impact assessment, a detailed list of mitigation actions is identified.



Types of impacts discussed earlier

Example



The Ahmedabad-Mumbai bullet train project will have various environmental, economic and social impacts that needs to be considered.

Source: economictimes.com

Environmental factors

- Land use and soil
  - Ex. Soil removal and pollution
- Land-cover change:
  - Ex. Wetlands, floodplains, mangroves, other subsoil (composition, depth, etc.)
- Surface
  - Ex. types and distribution, characteristics, uses, etc.
- Topography
  - Ex. Altitude, gradients, relief variations, orientation, etc.



Example: Mangroves of Sundarbans are highly valued eco-sensitive region in India

Environmental factors (cont'd..)

- Sewage release (treated & untreated)
  - Ex. Groundwater (location, description of aquifers, recharge areas, level of use, etc.)
- Surface water
  - Ex. location and description of areas that could be affected by the project
  - Identification of use of surface water; description of drainage areas and channels
  - Potential for floods, sedimentation, erosion, and eutrophication of water sources



Example: Untreated sewage contaminates the ground water or natural water resources

Environmental factors (cont'd..)

- Air pollution
  - Ex. Emissions such as noise, odor, dust
- Climate factors
  - Ex. rainfall, temperature, radiation, fog, wind, etc.
- Identification of sources of pollutants
- Fragile areas



Environmental factors for Impact Assessment (cont'd..)

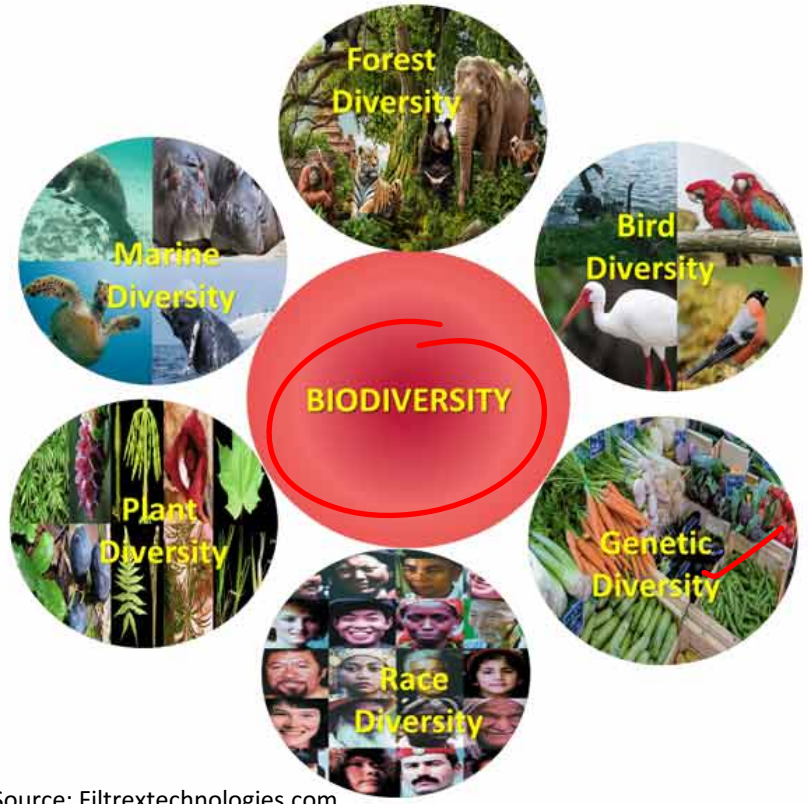


Image Source: Filtrextechnologies.com

- **Biodiversity:** impacts on endangered species
- Trees and canopy, migratory birds, indicator species and different groups of species and introduction of invasive species

Socio-cultural factors



Example: Natural life style and habitat of tribes on some Islands of Andaman & Nicobar have been preserved by not allowing external interventions.

Socio-economic factors

- Present land use and zoning
- Land-use plans
  - Ex. utilization or master plans, including the project area and surroundings
- Future trends or development pressures
- Resettlements



Example: Construction of Sardar Sarovar dam caused mass resettlement and there were agitations by several groups.

Historic & Heritage factors

- Community impacts
  - Ex. impact on poor, low-income people
- Impacts on culture or heritage
- Sites of potentially significant archaeological value
- Visual resources
  - Ex. physical description of the community



Example: Hazrat Ganj area of Lucknow is a special heritage locality which civil society wants to keep preserved.

Historic & Archaeological factors

- Natural areas of significant scenic value
- Historic and archaeological resources
  - Ex. listed historic areas or structures, according to national or community designation
- Other impacts
  - Ex. access to jobs, food, energy, aesthetics

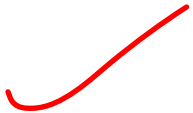


Example: Taj Mahal in the city of Agra.

Impact Assessment Parameters: World Bank









Example of Impact Assessment Parameters: World Bank

Environmental elements	Environmental impact factors
Social environment	Traffic, development of social economy
	Land use
	Demolition and resettlement, traffic, irrigation works, social economy
Living environment	Traffic, exchange, employment, income and safety
Ecological environment	Water and soil loss
water environment	Water pollution during construction period and operation period, accident risk
Acoustic environment	Traffic noise
Air quality	Raise dust、TSP
	Harmful substances in tail gas（NOx、CO）

Impact assessment table of world Bank for a town project.

Source: worldbank.org

Impact Assessment Approach: World Bank

- Findings of country specific environmental studies
- National environmental/climate change action plans; the country's overall policy framework, national legislation, and institutional capabilities related to the environment and social aspects
- Obligations of the country, pertaining to project activities, under relevant international environmental treaties and agreements.

Indian govt. has its National Action Plan for Climate Change (NAPCC) under which various missions National Solar Mission, National Water Mission, Green India Mission etc. are part of it.

Impact Assessment Approach: European Union (EU)

- Comparison of base line and alternatives.
- Assessment of environmental and wider socio-economic impacts in a qualitative and quantitative manner.
- Alternatives that have the same impacts can be group together and then separately the differences in impacts between the alternatives need to be highlighted.

EU is a political & economic union of 27 European countries with single currency and many common regulations & laws. It also facilitates many aid projects across many developing or under- developed nations.

Conclusion

- Environmental Impact Assessment is a Multi-stage process for identifying the potential externalities of a project, especially the negative impacts and suggesting possible mitigation measures.

- The Screening, scoping and the baseline studies collects sufficient data required to predict the potential impacts from a project.

- The Environmental and Socio-economic factors are considered while suggesting the mitigation measures for the project.

To be continued..

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Thank You